Trenton Banks

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EDUCATION

Cornell University : Ithaca, New York **GPA: 3.4/4.0**

Bachelor of Arts in Information Science and Economics; Minor in Computer Science and Business **June 2028**

•Relevant Coursework : Data Structures and Algorithms, Functional Programming, Probability and Statistics, Micro-Economics, Intermediate Macro Theory, Multivariable Calculus, Data Science in Python, Networks

PROFESSIONAL EXPERIENCE

Nuveen Investment Management **June - August 2026**

Software Engineering Intern **Charlotte, North Carolina**

•Built an agentic natural-language query tool for the fixed income team's internal dashboard, letting analysts retrieve live trade, volume, and position data through plain-English chat instead of manual SQL , delivering responses in under 15 seconds at 99% uptime and dramatically streamlining data access.

•Architected the system in Python with FastAPI, applying object-oriented design (abstraction, encapsulation, modular services) and formal architecture modeling to orchestrate a multi-tier pipeline across the presentation, application, and data layers.

•Developed a retrieval-augmented (RAG) text-to-SQL engine over a pg-vector store that converted questions into validated PostgreSQL queries against a read-only replica via the dashboard's API; authored test cases verifying API connectivity, data-retrieval accuracy, and SQL guardrail enforcement to ensure reliable, correct results.

CSMore(Cornell University) **July - August 2025**

Software Engineering Intern **Ithaca, New York**

•Designed a fully functional RISC-V processor in Logisim, designing datapaths, ALUs, control logic, and memory units aligned to ISA standards.

•Implemented and debugged software in C and OCaml across Docker and SSH environments, including an anagram detection algorithm using string manipulation, character frequency analysis, and hashmaps.

Chicago Transit Authority **June - August 2024**

Data Engineering Intern **Chicago, Illinois**

•Partnered with cross-functional engineers to analyze infrastructure workflows, applying lean principles to identify efficiencies and propose process improvements.

•Supported engineering teams on hands-on system design and optimization projects, building skills in process analysis.

Dawn Skies Therapeutic Interventions LLC **May - June 2024**

Software Engineering Intern **Chicago, Illinois**

•Engineered and deployed a responsive, front-end website using modern web technologies to establish the company's digital presence and improve user accessibility.

•Created HIPAA compliant intake forms that cut client onboarding time by 30% and ensured regulatory compliance.

TECHNICAL SKILLS

Programming Languages: Java, Python, Dart, JavaScript, C, OCaml, Assembly, C++, SQL, NodeJs, HTML/CSS

Frameworks and Runtimes: Kivy, Flutter, Node.js , FastAPI, NumPy, Pandas

Environments and Tools: VS Code, PostgreSQL, DBeaver, IntelliJ, Anaconda, Docker, AWS, Git/Gitlab, Kiro, Excel, Vite

PROJECTS

N.E.S.T **July 2026**

•Orchestrated a multi-agent AI system (AWS Bedrock, LangGraph, Claude) with an orchestrator agent routing queries to a deterministic engine for real-time financial calculation,deployed via a full-stack FastAPI and React application.

Pac-Man Mini Game **April 2025**

•Programmed an object-oriented Pac-Man-style game with AI pathfinding using a custom priority queue and Dijkstra's algorithm, integrating graph traversal and event-driven GUI logic across a 3,000+ line codebase using Java.